The Influence of Trends on Action: Analyzing the Public Sentiment and the Drivers of Peak Conversation Trends with Action Mobilization on the Grounds of the "17+8" Movement

Alvin Setiawan
Information Systems Department
School of Information Systems
Bina Nusantara University
Jakarta, Indonesia 11530
alvin.setiawan004@binus.ac.id

Fedric Winardo
Information Systems Department
School of Information Systems
Bina Nusantara University
Jakarta, Indonesia 11530
fedric.winardo@binus.ac.id

Christopher Reyner Gunawan
Information Systems Department
School of Information Systems
Bina Nusantara University
Jakarta, Indonesia 11530
christopher.gunawan005@binus.ac.id

Sulistyawankim
Information System Department
School of Information System
Bina Nusantara University
Jakarta, Indonesia 11530
sulistyawankim@binus.ac.id

Evaristus Didik Madyatmadja
Information System Department
School of Information System
Bina Nusantara University
Jakarta, Indonesia 11530
emadyatmadja@binus.edu

Abstract–The study of offline movement phenomenon that is started from viral topics on social media has not been a scope in sentiment analysis research. This article explores this dynamic through a case study of the "17+8 Movement," a definition of how heated online interaction can be a driving force for offline mobilization. While the connection between trends in digital conversation and real-world action seems to be well established, the processes that connect these two worlds remain poorly understood. Using a comparative methods approach that includes sentiment analysis and social network analysis, this study identifies the factors driving online-to-action transition. The findings of this study aim to contribute to the theory of social mobilization in the age of digital and practical implications for social activists, policy makers, and scholars.

Keywords–Mass Mobilization, Social Media, Public Sentiment, Collective Action

I. INTRODUCTION

Offline mobilization from social media debates and news has become a defining factor of political life in the digital era. Twitter, Instagram, and TikTok have become the new public squares in which collective opinions and emotions are shaped instantaneously and on a massive scale [1]. Such viral conversation trends, defined by a spike in the use of certain hashtags or keywords, can also help bring fringe issues to the forefront of public discussion. In the end, these trends frequently turn into a catalyst for offline actions in the real world.

Studies from the Reuters Institute Digital News Report 2024, shows that 27% of users report difficulty in distinguishing trustworthy news on TikTok, compared to 24% on X and only 13% on Google Search [18]. This can be summarized that TikTok awareness, X remains the discussion platform for verifiable political discourse and text-based movement coordination.

The "17+8 Movement" represents a significant context in which to examine the issue, since it consolidated 25 different demands; 17 short-term and 8 structural reforms turned into simple and concise requests to the Indonesian government that allowed it to attract substantial public attention [5]. It thus gives evidence of how online activity can be intense enough to spark mobilization offline, which is a link that was reflected in the positive correlation between digital use and civic participation found in more than 80% of case studies under Boulianne's meta-analysis [9]. Its broad resonance had been covered far and wide in mainstream media, with quite a few news reports relating an exponential development of online activity capitalizing on 57% of the population that now relies on social media to get breaking political news in the immediate run-up to the offline protests [18].

Sentiment analysis is a deeply understood method for social media opinion analysis [2], yet the process of social media conversations to being an offline participation is not well studied [3], [6], [7]. Especially on how sentiment category, dominant narratives, interactions, and influencers opinion impacts mass action. This lack of studies calls for a focused investigation for "17+8" demonstration, where this study tries to answer:

1. How can social media sentiments be visualized to determine patterns leading to real-world action?
2. What contributes to the media trend to bring critical points that leads to demonstration, and how do these correlate?
3. What determines real-world mobilization more; volume of digital conversation or intensity of negative sentiment?

To answer these questions, this research has four goals: first, to detect and analyze the public sentiment on social media towards the "17+8 Movement" through sentiment analysis and social network analysis; second, to identify the key factors that convert conversation trends into real-world mobilization using content analysis and interviews with participants; third, to examine the relationship between online conversation dynamics and offline actions, and to assess whether this relationship is causal; and fourth, to explore the function of key agents such as influencers or organizations in the process of translating digital trends into physical action.

This work contributes to the literature in two ways. Theoretically, it is expected to contribute to the communication and sociology fields by advancing the theory of social mobilization in the era of digital communication. It is also intended to contribute to a new analytical framework for quantifying the impact of online sentiment on collective behavior in the real world [9].

On a practical level, the implications of this study are to; answers on how to effectively mobilize through social media and digital communication [10] for activists. For the government, enhancing pattern recognition in public opinion understanding, then making effective communication strategies. For academic purposes, this study is intended for deeper analysis that may be used as a basis of research; including prediction models in data science and Artificial Intelligence[11]. In summary, "17+8 Movement" in this case will be a connection between sentiment analysis and collective offline action.

The study is not recommended to be used as foundational strict causality, but focuses on temporal correlation patterns between online social media sentiments and offline events.

II. LITERATURE REVIEW

A. Origins and Framing of the 17+8 Movement

The "17+8 Movement" emerged in late August 2025, while peaking in popularity in early June 2025; consisting of 25 demands from civilians to the government, this movement is popularized rapidly through social media platforms across the country.

Early scholarly accounts place the movement as a hybrid outcome of decentralized civil society groups, social media celebrities, and student networks that merged over 200 civil society demands into an effective platform that later became a trigger for countrywide protests. Recent Indonesian journal research narrates how the movement's purpose was all about institutional change, accountability, and openness, and they chart how that framing was deliberately designed to be shareable on social media and actionable on street protests [12], [13].

B. Digital Formation of the Movement

Scholar work on new Indonesian mobilizations indicates that viral demands often follow a two-stage digital origin:

1. Simplification and aggregation of fragmented resentment into a dense, entertainment-friendly platform; and,
2. Magnification by high-reach accounts such as influencers, student unions, and community leaders that turn online interest into offline mobilization.

The 17+8 invitations were quickly shared and spread through mainstream internet publics and then re-shared by huge student and activist networks an information ecology replicated in quick academic studies of the protests [12] and complementary qualitative research on digital activism in Indonesia detailing how hashtags and short symbolic frames facilitate quick identification and coordination for youth participants [14], [15]. These productions illustrate the loop of online activism: composition (demanding building), symbolic branding (17+8), influencer seeding, and distributed grassroots sharing, each step intensifying the likelihood of offline participation.

C. Online to Offline Mobilization

Qualitative studies of Indonesian youth movements "#17plus8" and campaigns like it would uncover a range of mechanisms for translating online opinion into physical presence: narrative resonance (shared morals and stories), logistical coordination (calls to action, meeting places promoted on social accounts), and relational connections (friend networks and student groups that reduce the cost of participation). Applied to 17+8, rapid academic accounts and interviews suggest that social media were a multiplier of emotions as well as a utilitarian coordination tool; young activists employed social spaces to share safety procedures, organize campus coordinating centers, and crowd-source legal-aid contacts practices validated by qualitative interviews from recent research on Indonesian online activism [14], [16].

In total, the literature indicates that alone, online mobilization is not enough, but the 17+8 case reaffirms the trend that hard, emotionally loaded frames with complementing offline mobilization channels are a winning formula for triggering mass mobilization.

D. State Response and Communication

Rapid academic commentaries and a particular journal article examining crisis communication by the state during 17+8 protests record two connected dynamics: (1) official public communication and apology/clarification strategies altered the course of resentment online; and (2) observable state reactions such as negotiation, concession in some form, or violent dispersal looped back into movement narratives and recruitment.

Early peer-reviewed research indicates that government crisis communication in tone and channel selection was deemed significant in de-escalating or escalating the protests, and that the 17+8 incident indicates the importance of rapid, trustworthy public communication to avoid further escalation of misleading rumor and retaliatory escalation [12], [17]. The literature thus places 17+8 not just as an example of a digital-first mobilization, but as a live experiment in how crisis communications operate today to enable the online to offline feedback loop.

III. RESEARCH METHODOLOGY

The research adopts a qualitative research method facilitated by data analysis in order to investigate how online social media sentiment is used to mobilize offline during the case of the "17+8 Movement.".

While the movement utilized visual platforms (TikTok/Instagram) for amplification, X served as the primary source for text-centric platforms for news and online discussion, making it the optimal source for text-based sentiment analysis. According to the Reuters Institute Digital News Report, 57% of Twitter users trust the platform for breaking news and political discussion, compared to Tiktok that only has 22% of political discussion, which is dominated by entertainment content [18]. Furthermore, Indonesian X facilitates complex discussions and logistical planning, where the use of short-form video platforms function primarily as entertainment space with lower information density [19].

Information is gathered using web scraping methods via Twikit Library through Python to collect public tweets containing words, phrases, or discussions about the movement. Gathered data are username, tweet text, time stamps, replies, and URL of tweet, filtered for relevance and popularity. Ethical research practices are prioritized in the data treatment for analysis and collection process. To ensure the reliability of our sentiment classification, we employed a pre-trained transformer-based machine learning model; IndoBERT to categorize sentiments. This decision is taken based on research by Septiani where IndoBERT has demonstrated superior performance and achieved 84.94% accuracy compared to generic modules that only achieved <= 70% accuracy [4].

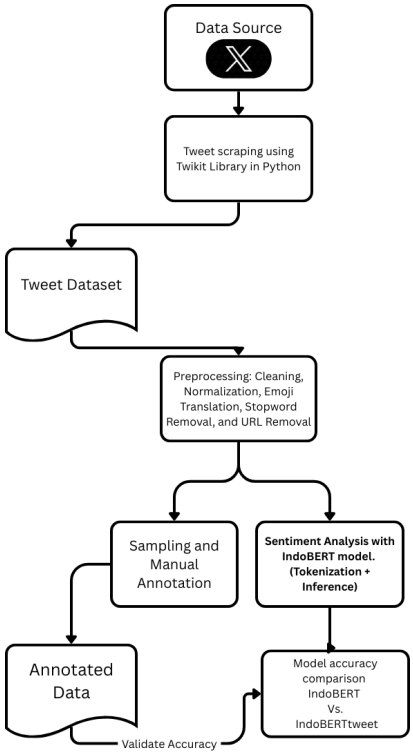


Fig. 1 Sentiment Analysis Overview Model

The method of sentiment analysis in this study starts from data collection from X platform, using the Twikit Python library to compile a raw tweet dataset. The data is then preprocessed to enhance input quality and increase prediction accuracy through noise reduction. Preprocessing includes essential text cleaning, normalization, removal of URLs, and translation of emoji. This preprocessing step is optimized for NLP algorithms to understand tokenized Bahasa Indonesia tweets in the dataset.

After preprocessing, the methodology then is splitted into parallel streams to benchmark model performance against ground truth. The tweets are then inputted using a pre-trained NLP algorithm; IndoBERT, with the model inputs optimized for the most accurate preprocessing method. Subsets of the dataset are also sampled with manual annotation as a baseline to determine prediction accuracy.

IV. RESULTS

A. Measurement

In order to measure public reaction to the “17 + 8 Movement”, a comparative framework was designed to determine the NLP model with the most accurate result. Comparative modelling approaches using the IndoBERT model determine the public opinion based on classification that are Positive, Neutral, and Negative. This allows for understanding complex public sentiment accurately where sarcasm or slang might otherwise be measured incorrectly.

The conversational 'texture' was also analyzed through more visual and statistical metrics. Counting in total the number of tweets and identifying "Top Words" for both positive and negative sentiments allowed us to see precisely which terms were driving the anger or hope. The significant words described in positive and negative sentiments were displayed in Word Clouds, which facilitates comparison between the vocabulary of solidarity and the vocabulary of frustration (e.g., 'Justice', 'Green' versus 'DPR', 'Police').

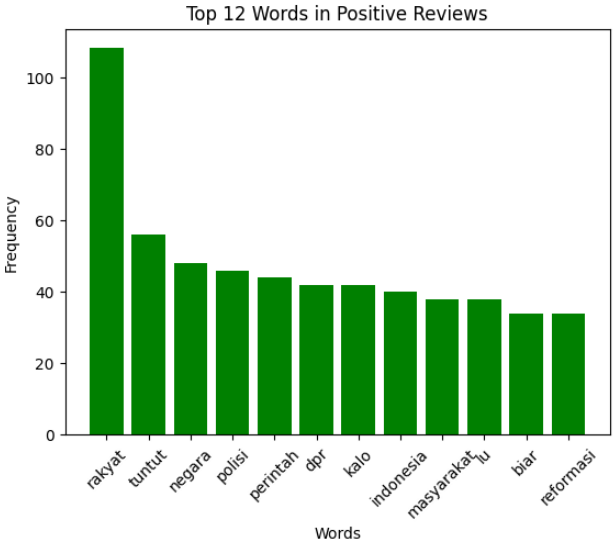
B. Data Collection Model

The data were obtained from the X platform (previously known as Twitter) by using a Python script developed with the Twikit library. To analyze the semantic texture, The Twikit library was an effective tool that enabled the collection of data from all related discussions involving the “17+8” demands and related protests, thereby ensuring that the data presented was raw and unfiltered. A total of 3.132 tweets were collected from August to October 2025 using hashtag and keyword filtering related to the 17+8 movement. Dataset consisting of tweets are then filtered with parameters such as; non-Indonesian languages, commercial advertisements, and mentions were removed; therefore resulting in 1.380 valid tweets for analysis.

The dataset was the backbone of our research. It was important that our dataset included the most important period of August to October 2025, encompassing the complete lifecycle of the phenomenon. Before conducting the analysis, our data was strictly implemented in the Preprocessing phase, including operations such as Cleaning (to remove noise), Normalization (to remove slang expressions), Tokenization (to split text into chunks), Stopword Removal (to remove irrelevant terms), and Link Removal (to remove linked items in X like images and videos). The IndoBERT model was intended to analyze pure, untainted text.

C. Data Analysis

The sentiment analysis was sharply divided based on the IndoBERT model. The text was able to categorize sharply based on the sentiment: “Negative” was predominant with 59.7% and tended towards the government failure and the police brutality, while “Positive” (14.3%) was not about “happiness” but about “solidarity” that is pictured in the frequency of word such as “rakyat” that means citizen as a whole and “tuntut” that describes solidarity to demand in terms of mutual support and in support of justice.



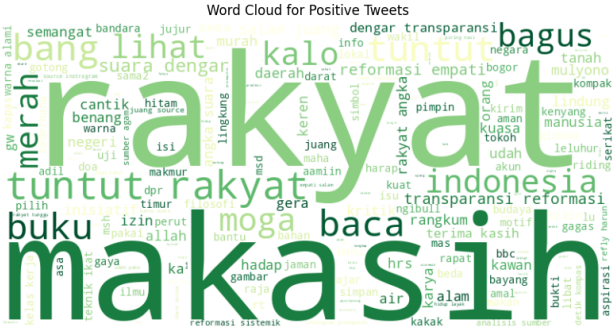


Figure 2. Positive Sentiment Bar Chart and Word Cloud

Sentiment analysis revealed Positive sentiment from Fig.2. Showing that classification of positive feeling does not correlate to happiness, but in **solidarity**. This interpretation is strongly supported by the word cloud, where one of the most frequent words is **"tuntut"** (demand/claim). The highest frequency of words is **“rakyat”** describing the collective message of citizen’s mutual support and demand for justice. Thus aligns with a socio-political response rather than an emotional one.

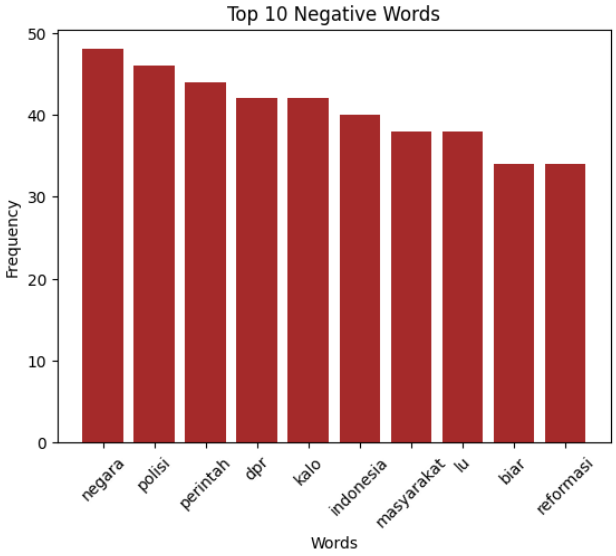




Figure 3. Negative Sentiment Bar Chart and Word Cloud

Fig. 3. Analysis of Dominant Negative Sentiments. Like the sentiments of negative tweets using frequency distribution in Bar Chart form and Word Cloud; the negative sentiments are once again not describing negative emotions, rather a disappointment in the country's institution. This data confirms that the “17+8” movement is a negatively structured movement. The high frequency of **“polisi” and “dpr”** indicates the public anger on official institutions in the country, furthermore this provides a location for abstract anger to manifest into offline movement.

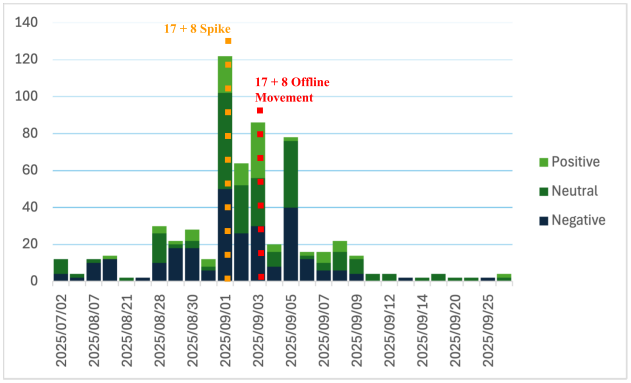


Figure 4. Temporal Correlation Between Online Sentiment Spikes (Sept 1) and Offline Demonstration (Sept 3), Demonstrating a 48-Hour Predictive Lag in the '17+8' Movement.

Figure 4 shows further evidence of relation from 17+8 online movement to offline events. Sharp rise in sentiments on the 1st of September fuels the offline demonstration on September 3rd. By tracking tweet sentiments, observation can be done to predict lag on online spike peaks approximately 48 hours prior to offline movement. The visualization reveals a specific pattern that is a 'precursor spike' in negative sentiment in government that results in physical action by exactly two days.

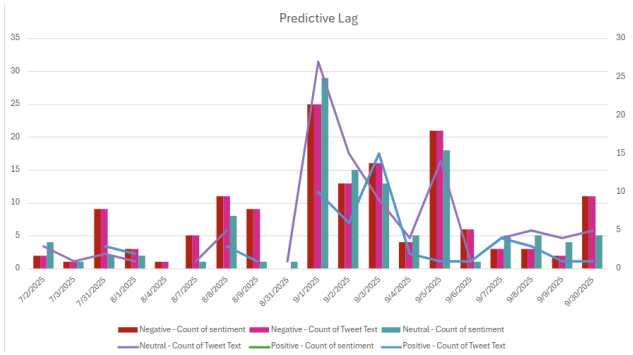


Figure 5. Predictive Lag from Negative Sentiment Tweets to Demonstration

Figure 5 yet again proved that, as hypothesized, the data reveals a clear "precursor pattern." While the offline demonstration occurred on September 3, 2025, the digital conversation did not peak simultaneously. Instead, a sharp, distinct spike in negative sentiment was observed on September 1, 2025, exactly 48 hours prior to the event. On Sept 1, negative tweet volume surged to its highest point (approx. 30 tweets/day), significantly outpacing neutral and positive discourse.

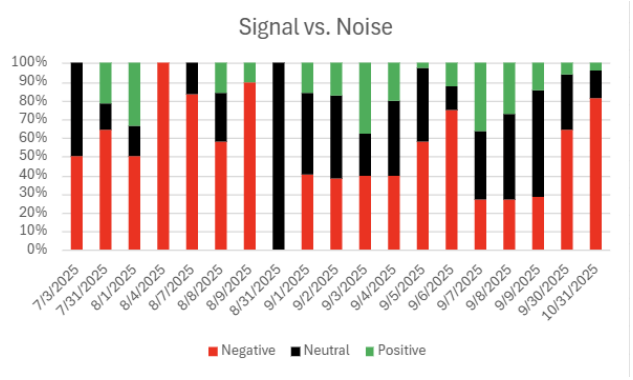


Figure 6. Sentiment Composition Analysis (Signal vs. Noise)

In figure 6, the analysis challenges the reliability of raw conversation volume as a primary mobilization predictor. As shown in the 100% stacked analysis, high tweet volume often correlates with "noise" (Neutral sentiment) rather than action. In contrast, the "signal" for mobilization is defined by sentiment composition. On the critical precursor date of **September 1**, Negative sentiment (Red) structurally displaced Neutral discourse (Black), signaling an active shift in public mood. Furthermore, dates with lower volume but near-100% negative saturation (e.g., Aug 4, Aug 31) indicate sustained momentum from core activists. This confirms that **sentiment intensity (polarity ratio)** is a significantly more precise predictor of real-world mobilization than aggregate tweet frequency.

The graph challenges the reliability of tweet volume as a demonstration predictor. This is shown with a stacked graph. This shows the comparison on volume of tweets compared to negative sentiments; where this is defined as being the "signal". Negative sentiment (Red) structurally displaced Neutral sentiment (Black), signaling an active shift in public mood. This indicates sustained momentum from core activists.

V. DISCUSSION

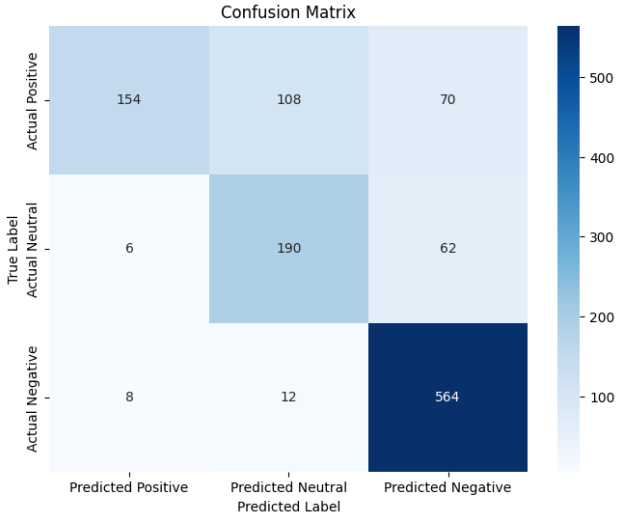


Fig. 7 Confusion Matrix of IndoBERT

A. IndoBERT Model Evaluation of Negative Sentiments

As visualized in the confusion matrix (Fig. 7), the IndoBERT model proved exceptionally reliable where it mattered; the Negative category. It correctly identified 564 negative tweets while misclassifying only 20, achieving a robust overall weighted precision of 79.70%.

Since the "17+8" mobilization was driven by the intensity of negative sentiment, the model's high sensitivity in this specific quadrant confirms that the critical spike observed on September 1st was a genuine warning signal, not a classification error.

B. Discussion Doesn't Mean Movement

Our analysis of the "17+8" gives a finding: a crowded discussion does not mean a movement is forming. This is proven from comparison on daily tweet activity in Signal vs. Noise graph. The indicator for precise prediction is not on how busy the discussion is, but how it changes the narrative. "Polarity ratio" or how negative the discussion is, becomes a more reliable metric than trending topics.

C. Prediction Lag

The predicted lag becomes a warning that forms between an exploding negative tactic. The online anger didn't explode into a physical protest immediately; but it gives a 48-hour lag period. One of the indicators that makes the prediction more accurate is how vocabulary changes to more negative sentiments. Negative words spiked in frequency, then into a physical march.

D. *"Technocratic Deflection" Only Makes Things Worse*

From the government it is hard to give impact to the sentiment analysis that shows this strategy; where the government uses complex economic explanations. This strategy did nothing to lower the negative trend line. This makes things worse, because of the technical jargon that only makes the crowd more disappointed because they are not served with rational explanation.

IV. CONCLUSION

A. *Theoretical Implications*

This study helps our understanding of social media sentiment by proving that digital anger could have consequences that are physical action. By using both IndoBERT and defining the best, we can prove that social media acts like a predictable timer to start a movement. This identification of "48-Hour Precursor Rule" becomes our main contribution. In addition, we demonstrated that intensity of sentiment gives a clearer picture to mobilization than only tweets trend.

B. *Implication for the Government*

Stop obsessing over volume. The drop in tweet volume doesn't mean a good indication. Effective de-escalation requires addressing the core of the issue rationally rather than trying to explain it away with technical language data.

C. *Limitations and Future Work*

We focused heavily on the platform X because it remains the hub for text-based political discourse. However, we know that younger demographics are increasingly organizing on visual platforms like TikTok. Future research needs to apply this same kind of sentiment tracking to video content to see if the "48-hour lag" is shorter on faster-paced video platforms. We also need to test this "Volume vs. Sentiment" hypothesis on different types of unrest to see if it holds true outside the Indonesian context

ACKNOWLEDGEMENT

All authors contributed to the conceptualization and design of this research. Alvin Setiawan, Christopher Reyner Gunawan, Fedric Winardo, and Sulistyawankim all performed material collection, data annotation, data sampling, and analysis. Data extraction and analysis is performed by Alvin Setiawan with the help of Daffa Tarigan. No funding is sourced for this research.

This study utilizes data that are collected publicly from the social media platform X (formerly Twitter) using the open-source Twikit Python Library. To facilitate further research, the data used in this research is available publicly, and is available from Zenodo at https://doi.org/10.5281/zenodo.17906665.

REFERENCES

[1] Hyvärinen, H., & Beck, R. (2021). "Emotions Trump Facts: The Role of Emotions in Social Media: A Literature Review," 114, 106553. https://doi.org/10.24251/HICSS.2018.226

[2] Sharma, P., & Jain, R. (2020). "A systematic review on sentiment analysis techniques for social media data." Procedia Computer Science, 167, pp. 1530-1540. https://doi.org/10.1007/978-981-15-0029-9_7

[3] Rodríguez-Ibánez, M., Castejón-Mateos, F, et al (2023). "A Review on Sentiment Analysis from Social Media Platforms" Expert Systems with Applications, 56(4), pp. 1297-1309. https://doi.org/10.1016/j.eswa.2023.119862

[4] Septiani, S., Putri, et al (2024) "Deep learning-based sentiment analysis for social media: A focus on multimodal and aspect-based approaches" Applied and Computational Engineering, 10(11), pp. 4363-4375. https://doi.org/10.54254/2755-2721/33/20230224

[5] Bestari N, Pl (2024) "Tuntutan Rakyat 17+8 Viral di Media Sosial, Begini Isinya" CNBC Indonesia, September. 01, 2025.https://www.cnbcindonesia.com/tech/20250901082334-37-663072/tuntutan-rakyat-17-8-viral-di-media-sosial-begini-isinya

[6] Tufekci, Z. (2017). Twitter and Tear Gas: The Power and Fragility of Networked Protest. Yale University Press. https://d-nb.info/124031910X/34

[7] Bennett, W. L., & Segerberg, A. (2012). "The logic of connective action: Digital media and the personalization of contentious politics." Information, Communication & Society, 15(5), 739–768. https://doi.org/10.1017/CBO9781139198752

[8] Selim, S, Elmas, T. et al (2023). "Understanding the spread of viral content: A case study of Twitter." WWW '23 Companion: Companion Proceedings of the ACM Web Conference 2023, 15(7), e0236168. https://doi.org/10.1145/3543873.3587373

[9] Boulianne, S. (2015). "Social media use and participation: A meta-analysis of current research." Information, Communication & Society, 18(5), 524–538. https://doi.org/10.1080/1369118X.2015.1008542

[10] Seneviratne, O., & Vithana, U. (2021). "Sentiment analysis on social media platforms for crisis management: A case study." International Journal of Information Management, 60, 102377. https://doi.org/10.5824/ajite.2025.01.003.x

[11] Wang, W., Li, J., & Xu, Y. (2021). "The Influence of Cognitive and Emotional Factors on Social Media Users' Information-Sharing Behaviours during Crises." MDPI, 11(2), 556. https://www.mdpi.com/2076-328X/14/6/495

[12] M. R. Fariza, "Government crisis communication in the 17 + 8 protests," 2025. Taylor & Francis online. Available: https://www.tandfonline.com/doi/full/10.1080/01296612.2025.2561245.

[13] K. A. Salsabila, "17+8 Tuntutan Rakyat: Krisis Kepercayaan dan [analysis]," Jurnal Ilmu dan Isu Indonesia (JIIC) article page, 2025. Available: https://jicnusantara.com/index.php/jiic/article/view/5079

[14] S. L. Wahyuningroem, "Youth political participation and digital movement in Indonesia: the case of #ReformasiDikorupsi and #TolakOmnibusLaw," F1000Research, 2024. Available (open access): https://f1000research.com/articles/12-543

[15] A. Masduki, "Digital activism for press freedom advocacy in post-authoritarian Indonesia," MDPI (open access), 2025. Available: https://www.mdpi.com/2673-5172/6/3/101?utm

[16] B. H. Nugroho, "Digital activism and youth participation in Indonesia: A qualitative study," Priviet Social Science Journal, 2025 article page and PDF. Available: https://journal.privietlab.org/index.php/PSSJ/article/view/603

[17] R. Ida, "Politics in Indonesia: democracy, social networks and youth participation," (Tandfonline commentary / journal), 2025. Available: https://doi.org/10.1080/23311886.2024.24320

[18] N. Newman, R. Fletcher, R. K. Nielsen, and others, "Reuters Institute Digital News Report 2024," *Reuters Institute for the Study of Journalism*, pp. 10–15, 2024. https://reutersinstitute.politics.ox.ac.uk/digital-news-report/2024

[19] Sumartias, Suwandi & Aries, Dwia & Pulubuhu, Tina & Sudarmono, & Adi, Achwan & Ratnasari, Eny. (2023). Democracy in the Indonesian Digital Public Sphere: Social Network Analysis of Twitter Users' Responses to the Issue of Nationalism Knowledge Test at the Corruption Eradication Commission (TWK-KPK). Jurnal Ilmu Sosial dan Ilmu Politik. 26. 240-257. https://doi.org/10.221